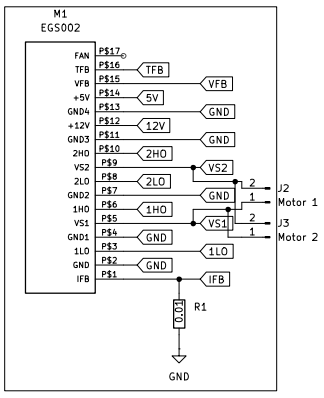
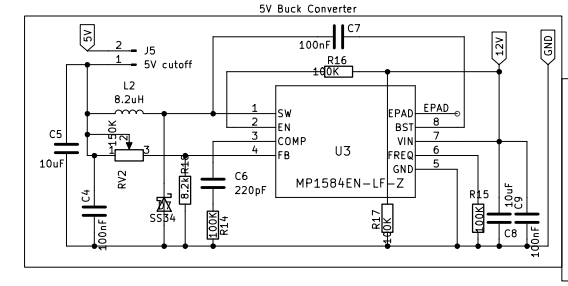
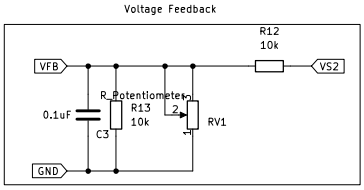
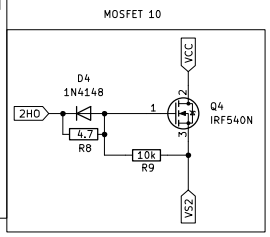
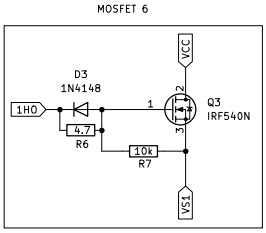
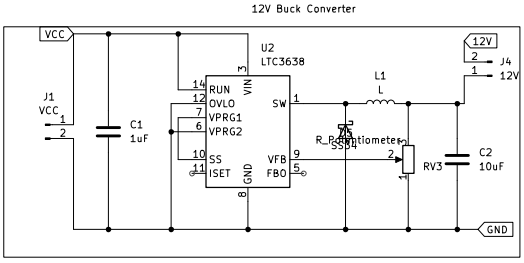

Main
M1
EGS002
FAN
TFB
VFB
+5V
GND4
+12V
GND3
2HO
VS2
2LO
GND2
1HO
VS1
GND1
1LO
GND
IFB
P$17
P$16
P$15
P$14
P$13
P$12
P$11
P$10
P$9
P$8
P$7
P$6
P$5
P$4
P$3
P$2
P$1
TFB
VFB
5V
GND
12V
GND
2HO
VS2
2LO
GND
1HO
VS1
GND
1LO
GND
IFB
J2
Motor 1
J3
Motor 2
R1
0.03
GND

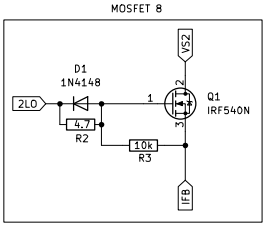

5V Buck Converter
5V
J5
5V cutoff
C7
100nF
R16
140K
12V
GND
L2
8.2uH
C5
10uF
150K
RV2
SS34
6.2R
C6
220pF
C4
100nF
100K
R14
U3
MP1584EN-LF-Z
SW
EN
COMP
FB
EPAD
BST
VIN
FREQ
GND
R17
140K
R15
100K
10uF
C9
C8
100nF

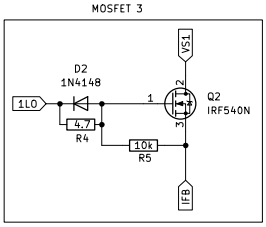

Voltage Feedback
R12
10k
VFB
VS2
R_Potentiometer
R13
10k
0.1uF
C3
RV1
GND


MOSFET 10
D4
1N4148
VCC
2HO
4.7
R8
10k
R9
Q4
IRF540N
VS2


MOSFET 6
D3
1N4148
VCC
1HO
4.7
R6
10k
R7
Q3
IRF540N
VS1


12V Buck Converter
VCC
J1
VCC
C1
1uF
U2
LTC3638
RUN
OVLO
VPRG1
VPRG2
SS
ISET
VIN
SW
VFB
FBO
GND
L1
L
R_Potentiometer
95k
RV3
C2
10uF
12V
J4
12V Cutoff
GND


MOSFET 8
D1
1N4148
VS2
2LO
4.7
R2
10k
R3
Q1
IRF540N
IFB


MOSFET 3
D2
1N4148
VS1
1LO
4.7
R4
10k
R5
Q2
IRF540N
IFB